

PUNTA ARENAS, Chile

WMO: 859340

Lat: 53.0033S

Lon: 70.8450W

Elev: 38

StdP: 100.87

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-5.0	-3.4	-7.9	1.9	-2.0	-6.4	2.2	-1.1	17.9	5.5	15.8	4.9	2.8	260	0.896

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.8	18.3	12.3	16.9	11.6	15.8	10.8	13.3	17.0	12.3	15.8	11.4	14.6	8.9	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.2	8.3	14.1	10.2	7.8	13.3	9.3	7.3	12.5	37.7	17.1	35.0	15.7	32.8	14.5	19.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	15.2	13.7	-8.4	22.5	2.2	2.2	-10.0	24.1	-11.3	25.5	-12.5	26.7	-14.1	28.3
			-8.9	15.5	2.1	1.4	-10.4	16.6	-11.6	17.4	-12.8	18.2	-14.3	19.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	10.8	10.5	9.1	6.7	4.2	2.0	1.9	2.8	4.7	6.6	8.3	9.9	(6)
(7)		DBStd	3.92	2.19	2.54	2.48	2.39	2.43	2.58	2.47	2.15	2.16	2.19	2.26	2.14	(7)
(8)		HDD10.0	1436	17	21	47	103	181	239	252	223	158	107	59	29	(8)
(9)		HDD18.3	4345	235	219	287	350	439	489	511	481	408	363	299	262	(9)
(10)		CDD10.0	133	40	35	18	3	0	0	0	0	0	2	9	25	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	1	1	1	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	6.6	7.6	6.9	6.5	6.2	5.6	5.4	5.7	6.0	6.3	7.2	7.8	7.7	(14)
(15)	Precipitation	PrecAvg	377	37	30	35	43	38	30	29	31	24	25	25	31	(15)
(16)		PrecMax	601	94	85	99	95	97	90	94	91	62	76	69	117	(16)
(17)		PrecMin	214	5	3	7	9	5	3	3	4	4	6	4	3	(17)
(18)		PrecStd	78	21	17	21	24	20	19	20	18	14	16	15	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.2	21.2	19.0	15.2	11.9	10.0	9.1	10.0	13.8	15.2	17.9	18.8	(19)
(20)			MCWB	13.3	14.5	12.6	11.4	8.9	7.9	6.4	6.6	8.4	9.1	11.0	12.0	(20)
(21)		2%	DB	18.0	18.6	16.2	13.1	10.0	8.0	7.2	8.2	11.1	13.2	15.4	16.9	(21)
(22)			MCWB	11.9	13.2	11.5	10.0	7.6	6.1	5.2	5.5	7.0	8.2	10.1	11.0	(22)
(23)		5%	DB	16.2	16.3	14.8	11.9	8.9	6.8	6.1	7.1	9.9	12.1	14.0	15.2	(23)
(24)			MCWB	11.1	11.7	10.6	9.0	6.9	5.4	4.5	4.9	6.3	7.5	9.1	10.1	(24)
(25)		10%	DB	15.0	14.9	13.1	10.4	7.9	5.8	5.2	6.1	8.5	10.9	12.8	14.0	(25)
(26)			MCWB	10.3	10.6	9.6	8.1	6.3	4.7	3.8	4.2	5.5	6.9	8.3	9.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.2	15.7	13.8	12.4	9.5	8.2	6.8	7.2	8.9	10.3	12.2	13.1	(27)
(28)			MCD B	18.7	19.7	17.5	14.1	10.9	9.6	8.2	8.9	13.1	13.9	16.1	17.1	(28)
(29)		2%	WB	12.8	13.7	12.1	10.6	8.0	6.3	5.5	5.9	7.4	9.0	10.7	11.8	(29)
(30)			MCD B	16.6	17.6	15.7	12.6	9.3	7.2	6.6	7.4	10.2	12.7	14.3	15.5	(30)
(31)		5%	WB	11.8	12.3	11.0	9.5	7.1	5.4	4.7	5.1	6.5	8.0	9.7	10.7	(31)
(32)			MCD B	15.4	15.7	13.7	11.3	8.1	6.2	5.7	6.4	8.8	10.9	13.0	14.2	(32)
(33)		10%	WB	10.8	11.1	10.2	8.5	6.4	4.6	3.9	4.3	5.7	7.1	8.8	9.9	(33)
(34)			MCD B	14.1	14.1	12.5	10.0	7.3	5.4	4.9	5.6	7.7	10.0	12.0	13.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	7.8	7.3	6.4	5.6	5.0	5.0	5.5	6.6	7.6	7.8	7.9	(35)
(36)			MCDBR	9.4	9.9	9.1	7.9	6.4	5.5	5.3	6.5	8.5	9.2	9.5	9.4	(36)
(37)			MCWBR	5.4	6.0	5.7	5.6	4.9	4.5	4.2	4.6	5.4	5.4	5.4	5.4	(37)
(38)		5% WB	MCD BR	9.0	9.3	8.2	7.4	5.6	5.1	5.0	5.9	7.5	8.6	8.9	8.9	(38)
(39)			MCWBR	5.7	6.2	5.7	5.8	4.7	4.5	4.3	4.6	5.2	5.5	5.6	5.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.325	0.318	0.319	0.310	0.290	0.278	0.288	0.301	0.316	0.319	0.322	0.325	(40)	
(41)		taud	2.548	2.575	2.558	2.544	2.489	2.391	2.436	2.518	2.509	2.518	2.522	2.531	(41)	
(42)		Ebn at Noon	980	951	886	788	682	610	666	785	872	939	977	989	(42)	
(43)		Edh at Noon	100	90	80	64	49	44	50	64	82	95	102	104	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.36	4.40	2.96	1.70	0.91	0.58	0.74	1.40	2.61	4.13	5.18	5.68	(44)	
(45)		RadStd	0.32	0.25	0.18	0.10	0.08	0.05	0.05	0.10	0.18	0.20	0.29	0.35	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	+0.62	N/A	N/A	N/A	-0.75	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	+0.75	N/A	N/A	N/A	-0.40	-84	N/A	N/A	N/A

Nomenclature: See separate page